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EEE444 Robust Feedback Theory - LabAssignment 1- Preliminary Work

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"I completed this homework assignment independently."

Part 1 a and b)

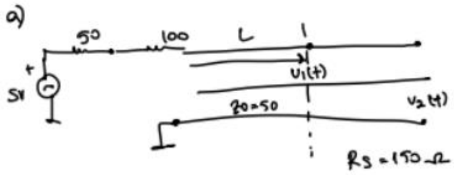
$V_c/C = 0.66 = v_p/c$ $C = 3.00 \times 10^8$ m/s

$V_c = v_p = 0.66 \times 3 \times 10^8 \approx 1.98 \times 10^8$ m/s

1sn 1.98×10^8 m

? 5m

a)



$\tau = \underline{\underline{25.25\,ns}}$ (for 5m), for 10m = 2L → 50.5ns

(from now on, we call τ = 25.25ns)

$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ when $Z_L = \infty$

$\Gamma_L = \frac{\infty - Z_0}{\infty + Z_0} = 1$ (open circuit)

(load end reflection)

$R_S = 150\,\Omega$

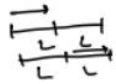
$V^+ = \frac{5V \cdot Z_0}{Z_0 + R_S} = \frac{5 \cdot 50}{200} = 1.25V$

$\Gamma_S = \frac{R_S - Z_0}{R_S + Z_0} = \frac{150 - 50}{150 + 50} = 0.5$

(when a backward wave returns, the amount of re-enters the line)

for $V_2(t)$

when $t < 2\tau$ $V_2 = 0$ (for the length of 2L = 10m)



when $2\tau < t < 6\tau$ $V_2 = (1.25V)2 = 2.5V$

" $6\tau < t < 10\tau$ $V_2 = 2(1.25 + \underset{0.5 \times 1.25}{0.625}) = 3.75V$

when $10\tau < t < 14\tau$ $V_2 = 2(1.25 + 0.625 + \underset{0.5 \cdot 0.625}{0.3125}) = 4.375V$

when $\underset{353.5ns}{14\tau} < t < \underset{404ns}{18\tau}$ $V_2 = 2(1.25 + 0.625 + 0.3125 + \underset{0.3125 \cdot 0.5}{0.15625}) = 4.6875V$ ⋮ continues like that

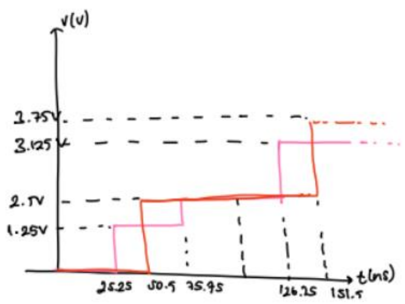
when $t \to \infty$ approaches to 5V

when $t \to \infty$ approaches to 5V

function of time (voltages of $V_1(t)$ and $V_2(t)$)

$V_2(t) = \frac{1 - \Gamma_S^{(\lfloor t - T)/2T)}}{1 - \Gamma_S}(1.25V) \cdot u(t - T)$

$V_1(t) = \frac{V_2(t + T/2)}{2} + \frac{V_2(t - T/2)}{2}$



for $V_1(t)$

first $\tau \to L = 5m$

then $3\tau \to 10m$, 5m returns back

$t < \tau$ $V_1 = 0$

when $0 < t < 3\tau$ $V_1 = 1.25V$

when $3\tau < t < 5\tau$ $V_1 = 1.25V + \underset{\text{reflected}}{1.25V} = 2.5V$

" $5\tau < t < 7\tau$ $V_1 = 2.50 + \underset{\text{reflected}}{0.625} = 3.125V$

" $7\tau < t < 9\tau$ $V_1 = 3.125 + 0.625 = 3.75V$

" $8\tau < t < 11\tau$ $V_1 = 3.75V + \underset{\text{reflected}}{0.3125} = 4.0625V$

" $11\tau < t < 13\tau$ $V_1 = 4.0625V + \underset{\text{refl.}}{0.15625V} = 4.21875V$

continues till $t \to \infty$ → reaches 5V

1) b) for $v_1(t)$:

at $t = t_\kappa$ and $t = 3t_\kappa \rightarrow 0.5^0 \times 1.25 = 1.25V$

" $t = 5t_\kappa$ and $t = 7t_\kappa \rightarrow 0.5 \times 1.25 = 0.625V$

" $t = 9t_\kappa$ and $t = 11t_\kappa \rightarrow 0.5^2 \times 1.25V = 0.3125V$

for $v_2(t)$:

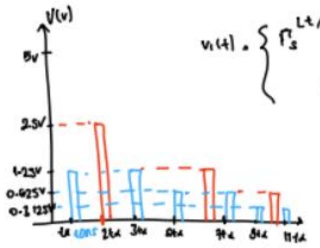
at $t = 2t_\kappa \xrightarrow{T} 1.25V \times (1+\Gamma_L) = 1.25 \times 2 = 2.5V$

at $t = 4t_\kappa \rightarrow 1.25V \times \Gamma_S = 1.25 \times 0.5 = 0.625$

$2T \rightarrow 0.625 \times (1+\Gamma) = 0.625 \times 2 = 1.25V$

at $t = 6t_\kappa \rightarrow 0.625 \times \Gamma_S \times (1+\Gamma_L) = 0.5 \times 0.625 \times 2 = 0.625V$

$3T$ ... goes like that so, the corresponding plot will look like this,



$$v_1(t) = \begin{cases} \Gamma_S^{\lfloor t/2T \rfloor} \times 1.25V & \text{when } t_\kappa < t \pmod{T} < t_\kappa + 10ns \\ 0 & \text{otherwise} \end{cases}$$

$$v_2(t) = \begin{cases} \Gamma_S^{\lfloor t/2T \rfloor} \times (V/2) & \text{when } T < t \pmod{2T} < T + 10ns \\ 0 & \text{othw.} \end{cases}$$

Figure 1: Part 1 Calculations and Results

Part 2

In this part I picked 280MHz, and accordingly all calculations below are made depending on this value:

2) for Z_1

$\beta = \frac{\omega}{v_p} = \frac{2\pi}{\lambda} \rightarrow \beta l$ = in radians (w)

phase velocity

$E = 360 \cdot \frac{l}{\lambda_{line}}$ $\quad \lambda_{line} = \frac{vf \cdot c}{f}$

$E = 360 \cdot \frac{88 \times 10^{-3}}{0.7948}$ $\quad \lambda_{line} \approx \frac{0.7 \times 3.00 \times 10^8}{280 \times 10^6} \approx 0.74948$

280MHz

$\approx 42.27°$

for Z_0: $\quad c \approx 3.00 \times 10^8$

$E = 360° \frac{l}{\lambda_{line}}$ $\quad \lambda_{line} = \frac{\overbrace{vf \cdot c}^{v_p}}{f}$ $\quad \beta = \frac{\omega}{v_p} = \frac{2\pi}{\lambda}$

phase velocity

$\cancel{2\pi} v_p = \lambda \omega \rightarrow \cancel{2\pi} f$

$v_p = \lambda f$

$\lambda = \frac{v_p}{f}$

$\lambda_{Z_0} = 0.8c / f \approx \frac{0.8 \times 3.00 \times 10^8}{280 \times 10^6} = 0.8571$

$E = 360 \cdot \frac{50 \times 10^{-3}}{0.8571} \approx 21.01$

Figure 2: Derivations of triple strub Matching

After these values are inserted, necessary tuning operations are made according to what we are asked.

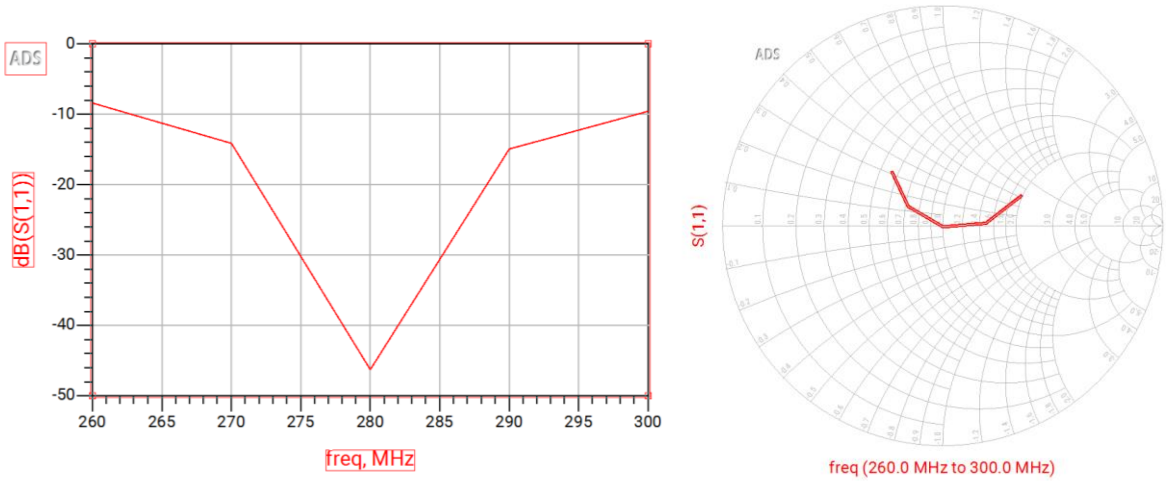


Figure 3: Tuned S11 Result at 280MHz

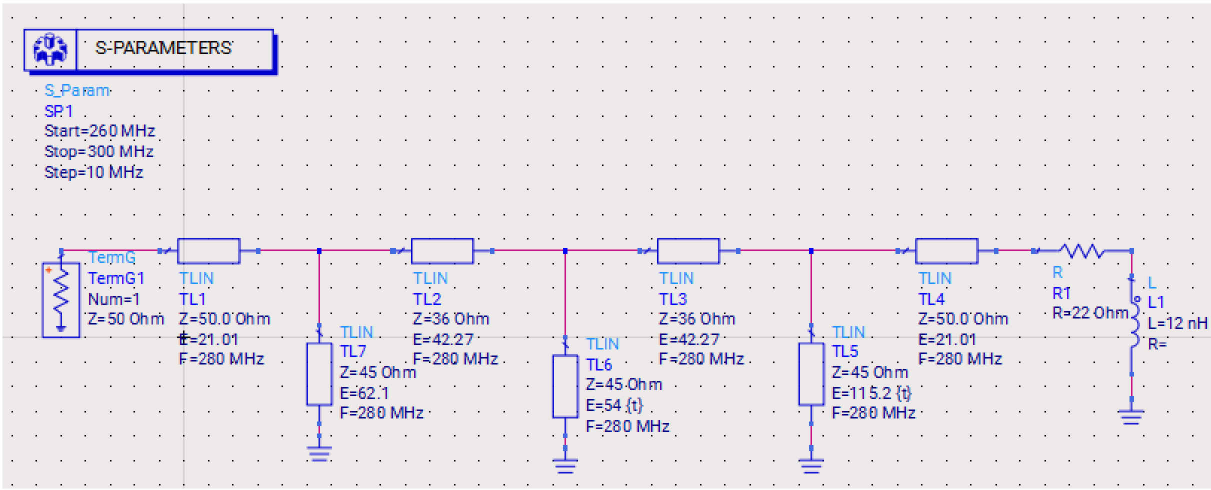


Figure 4: Tripl- strub Matching Circuit

The resulting S_{11} curve shows a deep notch at 280 MHz, which corresponds to nearly perfect matching. This confirms that the network achieves an excellent conjugate match between the 50 Ω source and the complex load. The Smith chart verifies that the impedance trajectory passes precisely through the chart center at 280 MHz, indicating zero reflection. All fixed and stub line lengths were adjusted directly in ADS to minimize S_{11} at 280 MHz, without using ideal lumped elements or theoretical simplifications.

Part 3)

a) We are given an original microstrip LPF design.T he dimensions of the microstrip lines are given in the schematic.We are asked to enter the schematic in ADS and plot the S11 and S21 of the filter.

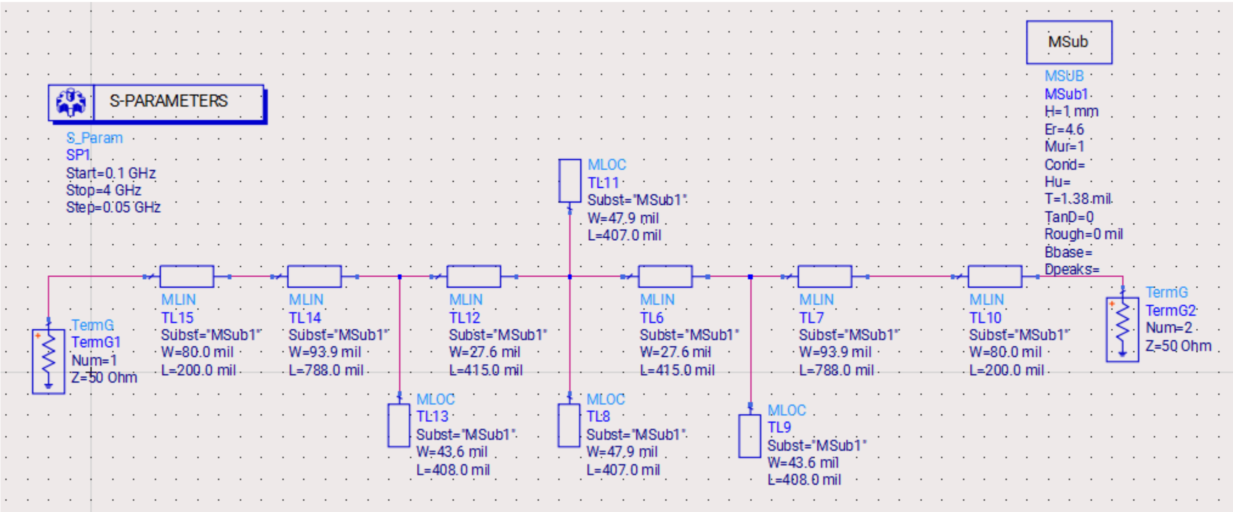


Figure 5: ADS Schematic of the Microstrip Low-pass Filter

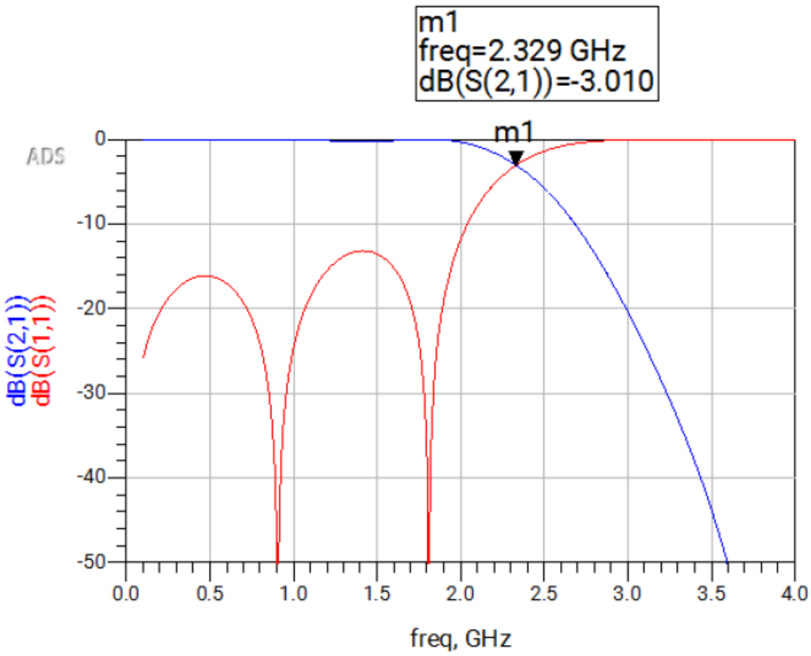


Figure 6: Microstrip Low-pass Filter Response (S11-S21)

b) In this part we modified the filter so that the corner frequency is different = 2.5GHz. Below calculations are made to find new lengths.

3)b) 2.3 – 2.35 GHz corner freq! $= f_c^{old}$

I pick $f_c^{new} = 2.5\,GHz$

$$S = \frac{2.33}{2.5} = 0.932$$

for TL15 → 200 × 0.932 = 186.4 mil
" TL14 → 788 × 0.932 = 734.416 mil
" TL13 → 408 × 0.932 = 380.256 "
" TL12 → 415 × 0.932 = 386.78 "
" TL11 → 407 × 0.932 = 379.324 "
" TL9 → 408 × 0.932 = 380.256 "
" TL8 → 407 × 0.932 = 379.324 "
" TL7 → 788 × 0.932 = 734.416 "
" TL10 → 200 × 0.932 = 186.4 "
" TL6 → 415 × 0.932 = 386.78 "

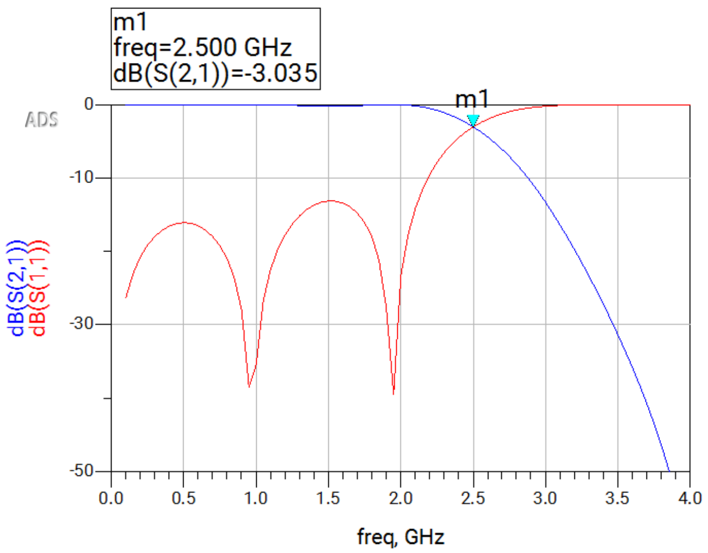


Figure 7: Microstrip Low-pass Filter Response redesigned when $f_c \approx 2.5GHz$ (S11-S21)

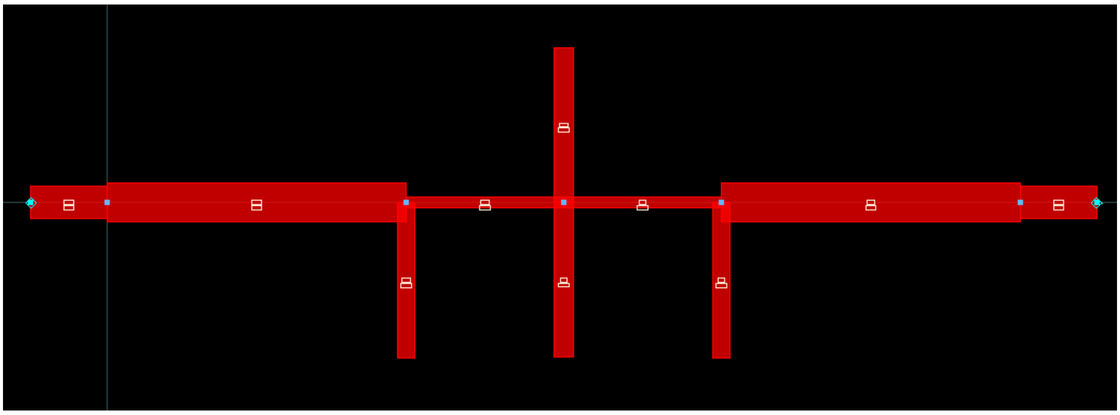


Figure 8: ADS Layout of the Low-pass Filter

Part 4)

a)

We have a microstrip band-pass filter built on the same substrate. Using the given schematic, we are asked to plot the S11 and S21 of the filter.

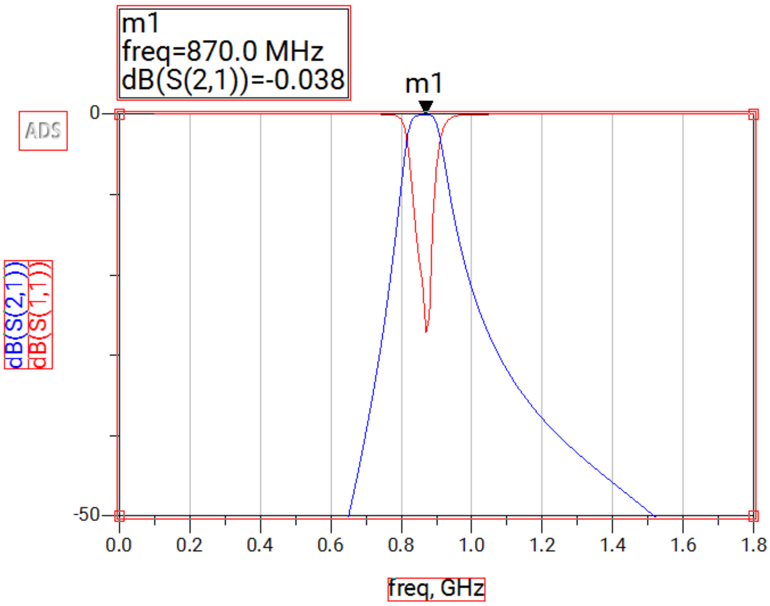


Figure 9: Microstrip BP Filter Response when $f_c \approx$ 870MHz (S11-S21)

b) In this part we modified the filter so that the corner frequency is different = 750M Hz. Below calculations are made to find new lengths.

4) b) $f_c^{old} = 870\ MHz$

$f_c^{new} = 750\ MHz$

$$S = \frac{f_c^{old}}{f_c^{new}} = \frac{870MHz}{750MHz} = 1.16$$

for TL27 → 468.29 × 1.16 = 543.2164 mil

" TL26 → 100 × 1.16 = 116 mil

" TL22 → 523 × 1.16 = 606.68 "

" TL20 → 150 × 1.16 = 174 mil

" TL19 → 150 × 1.16 = 174 mil

" TL18 → 400 × 1.16 = 464 "

" TL17 → 100 × 1.16 = 116 mil

" TL16 → 879.52 × 1.16 = 1020.2432 "

for TL28 → 468.29 × 1.16 = 543.2164 mil

for TL24 → 100 × 1.16 = 116 mil

" TL23 → 523 × 1.16 = 606.68 "

" TL21 = 150 × 1.16 = 174 "

" TL10 = 700 × 1.16 = 812 "

" TL12 = 700 × 1.16 = 812 "

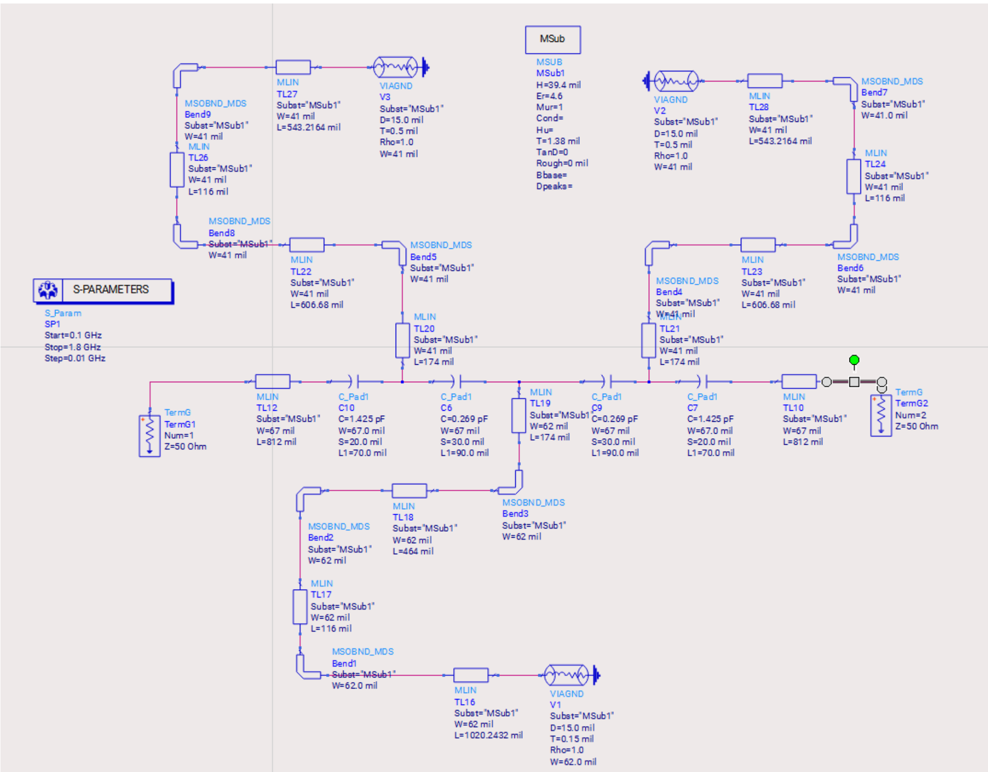


Figure 10: ADS Schematic After Scaling for $f_c = 750MHz$

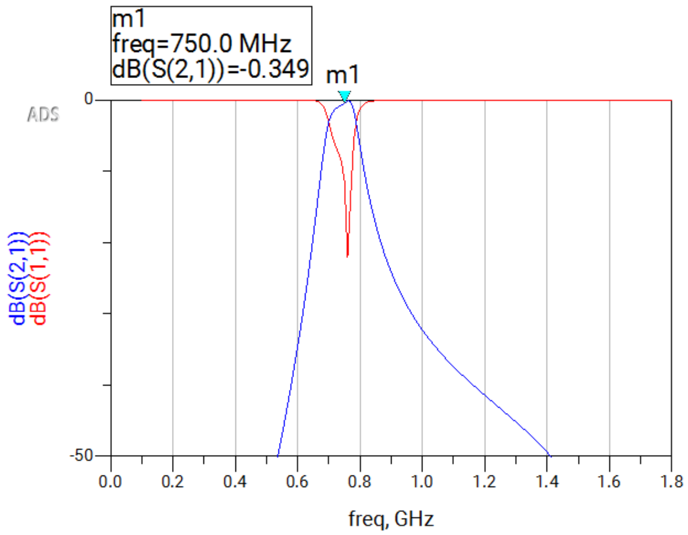


Figure 11: Scaled BPF Response When New fc= 750MHz

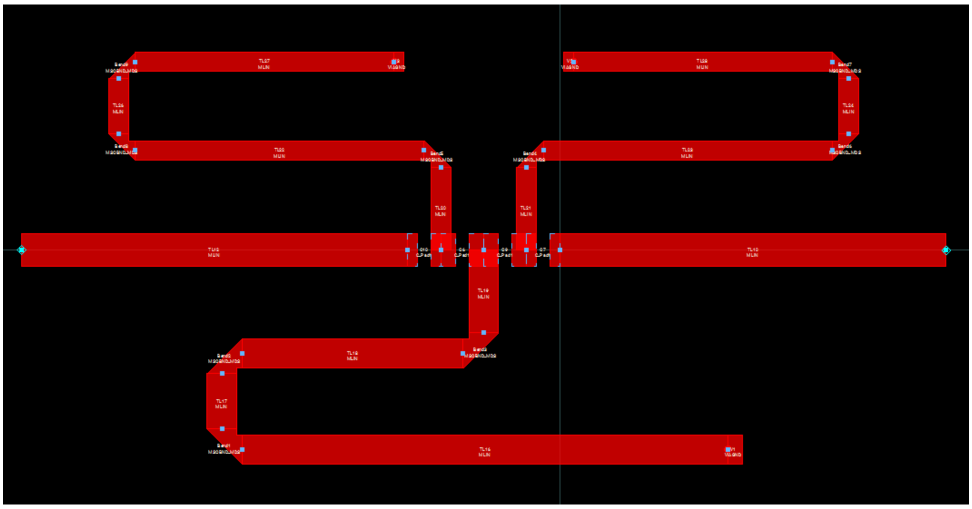


Figure 12: ADS Layout of the Band-pass Filter